Leptospirosis

21

Introduction

Zoonosis with protean clinical manifestations caused by pathogenic spirochetes of the genus *Leptospira*.

Synonyms: Weil's disease, Weil-Vasiliev disease, Swineherd's disease, rice-field fever, waterborne fever, nanukayami fever, cane-cutter fever, swamp fever, mud fever, Stuttgart disease, and Canicola fever.

Epidemiology

- ☑ It is distributed worldwide. Under-reported infection, there is no reliable global incidence data.
- ☑ Estimated 1.03 million cases worldwide annually with 58,900 deaths and loss of 2.9 million disability-adjusted life years.
- ☑ Leptospires survive longer in warm and humid conditions.
- ☑ Majority of clinical cases occur in the tropics.

Infection in animals:

- ☑ The organism infects a variety of wild and domestic mammals, especially rodents, cattle, swine, dogs, horses, sheep, and goats.
- ☑ Animals can be asymptomatic or develop clinical infection, which can be fatal.
- ☑ Reservoir animals may shed the organism in their urine intermittently or continuously throughout life, resulting in contamination of the environment, particularly water.

Epidemiology

Infection in humans:

- ☑ Humans most often become infected after exposure to environmental sources, such as animal urine, contaminated water or soil, or infected animal tissue through cuts or abraded skin, mucous membranes, or conjunctiva.
- ☑ Incubation period of 2–26 days (average 10 days).
- ☑ In the tropics, endemic leptospirosis is mainly a disease of poverty (including low education, poor housing, absence of sanitation, and poor income).
- ☑ Acquired through occupational exposure (subsistence farming) and living in rodent-infested, flood-prone, overcrowded, and water-logged urban areas.
- ☑ Large outbreaks affecting thousands of people and leading to hundreds of deaths are common occurrences, often associated with increased rainfall or flooding, which presumably increased the risk of exposure to contaminated water.

Anicteric Form

- ☑ The clinical course of leptospirosis is variable. Most cases are mild and self-limited or subclinical, while some are severe and potentially fatal.
- ☑ Clinically apparent leptospirosis presents with the abrupt onset of fever, rigors, myalgia, and headache in 75–100% of patients.
- ☑ Conjunctival suffusion in a patient with a nonspecific febrile illness should raise suspicion for the diagnosis of leptospirosis.

Other manifestations include:

- ☑ Meningoencephalitis where presentation is severe headache and neck stiffness.
- ☑ Myocarditis common in infants and toddlers can present as fever with disproportionate tachycardia, muffled heart sounds, and cardiomegaly.
- ☑ Polyserositis as pleural effusion, pericardial fluid, ascites, and gallbladder wall edema.
- ☑ Rarely as hepatorenal syndrome, pulmonary hemorrhage, acute respiratory distress syndrome (ARDS), uveitis, optic neuritis, peripheral neuropathy, and rhabdomyolysis.

Icteric Form (Weil's Syndrome)

Leptospirosis may be complicated by jaundice and renal failure (Weil's disease), and the disease closely mimics acute viral hepatitis and the differentiating feature may be presence of polyserositis and cholecystitis.

It is severe form of infection, less common in children but associated with higher mortality.

General Clinical Laboratory Findings

- ☑ Erythrocyte sedimentation rate (ESR) is elevated, white blood cell (WBC) counts range from below normal to moderately elevated.
- ☑ *Liver functions tests*: Elevated in aminotransferase, bilirubin and alkaline phosphatase, hyperbilirubinemia is out of proportion to jaundice in cases of icteric leptospirosis.
- ☑ *Renal function tests*: May be impaired, indicated by raised plasma creatinine.
- ☑ *Urine analysis*: Proteinuria, pyuria, microscopic hematuria, hyaline, and granular casts.
- ☑ *Lumbar puncture*: Elevated cerebrospinal fluid (CSF) pressure, predominance of lymphocytes and polymorphs.
- ☑ *Peripheral blood smear*: Leukocytosis with shift to left and thrombocytopenia.

Serologic Tests

Serologic testing is mostly used for diagnosis.

Enzyme-linked Immunosorbent Assay

- ☑ Useful for detection of IgM antibodies (which is positive from the 5th day of illness) and IgM-specific dot-enzyme-linked immunosorbent assay (ELISA) tests are now recommended in clinical practice.
- ☑ These tests have sensitivity >80–90% and are done at many regular pathological and microbiological laboratories.
- ☑ Between two- and fourfold rise in titer is suggestive of leptospirosis.
- ☑ Single high titer is usually seen during the 2nd or 3rd week of illness.
- ☑ The slide agglutination method, Dri-Dot assay, Lepto-Dipstick, latex agglutination, complement fixation assay, indirect immunofluorescent test, and indirect hemagglutination test are also available; these tests too have good sensitivity of up to 85%.
- ☑ *Microscopy*: Direct microscopic observation is used to detect leptospires in body fluids, check culture growths, etc.

Microscopic Agglutination Test

- ☑ It detects serovar-specific antibodies.
- ☑ Microscopic agglutination test (MAT) is the gold standard of the serodiagnosis for leptospirosis, because this assay has a high sensitivity and allows for the detection of group-specific antibodies.
- ☑ Two major disadvantages of this test are that in regions where leptospirosis is common, there may be a substantial proportion of the population with elevated titers of MAT and secondly, the performance of MAT is restricted to laboratories that are capable of maintaining strains for the preparation of live antigens.
- ☑ Therefore, serological tests remain suboptimal for clinical use in diagnosing leptospirosis.

Diagnosis

Other Tests

- ☑ *Dark-field microscopy (DFM):*
  - Leptospires are seen as thin, bright, actively motile rods, moving with characteristic rapid spinning and jerking motility.
  - DFM lacks sensitivity and specificity.
  - The positivity of DFM decreases from 100 to 90.9% with increase in the duration of infection for >1 week.
  - Another disadvantage of this technique is that both false-positive and false-negative diagnosis can be easily made even in experienced hands.
- ☑ *Histochemical stains and immunostaining:* It may be used to find leptospires where they are scarce, or where there is material that precludes the use of DFM.
- ☑ *Polymerase chain reaction (PCR):*
  - Molecular techniques to detect the presence of leptospiral deoxyribonucleic acid (DNA) in blood, urine, or spinal fluid have shown to be sensitive and specific.
  - Sensitive assay for the detection of *Leptospira* DNA that is based upon amplification of the *Leptospira rrs (16S)* gene have been developed.
  - PCR assay can be used on biological samples such as CSF, urine, or blood as a diagnostic tool for cases of suspected leptospirosis.
  - PCR cannot identify the infecting serovar.

Modified Faine's Criteria (Table 1)

Modified Faine's criteria with amendment in 2012 are very useful in diagnosis of presumptive cases of leptospirosis awaiting confirmatory laboratory reports. It includes clinical, epidemiological, and laboratory criteria with assigned individual scores.

TABLE 1: Modified Faine's criteria (2012).

Part A: Clinical data	
Question	*Score*
Headache	2
Fever	2
Temperature >39°C	2
Conjunctival suffusion	4
Meningism	4
Muscle pain	4
Conjunctival suffusion	
+ Meningism	10
+ Muscle pain	
Jaundice	1
Albuminuria/nitrogen retention	2
Hemoptysis/dyspnea	2
Total score	
Part B: Epidemiological factors	
Question	*Score*
Rainfall	5
Contact with contaminated environment	4
Animal contact	1
Total score	10
Part C: Bacteriological and laboratory findings	
Isolation of *Leptospira* in culture—diagnosis certain PCR	25
Positive serology	*Score*
ELISA IgM positive	15
SAT—Positive*	15
MAT—Single high titer+	15
Rising titer (paired sera)	15
Other rapid tests**	15

*Any one of the tests only should be scored 25.
**Latex agglutination test/Lepto Dipstick/LeptoTek Lateral Flow/LeptoTek Dri-Dot Test
Presumptive diagnosis of leptospirosis is made of Part A or Part A + B: 26 or more
Parts A, B, C (Total): 25 or more
A score between 20 and 25 suggests leptospirosis as a possible diagnosis.
(ELISA: enzyme-linked immunosorbent assay; IgM: immunoglobulin M; MAT: microscopic agglutination test; SAT: standard agglutination test)

Treatment

Stable patients: Any one of the following antibiotics is effective:

Drug	Dose	Duration	Route
Amoxicillin	50–100 mg/kg/day	7 days	Oral
Azithromycin	10 mg/kg/day	3 days	Oral
Cefixime	10 mg/kg/day	7 days	Oral
Doxycycline	5–10 mg/kg/day	7 days	Oral

Sick children: Meningoencephalitis, myocarditis, and renal involvement

Drug	Dose	Duration	Route
Penicillin	50,000–100,000 U/kg/day	5–7 days	IV
Ceftriaxone	75–100 mg/kg/day	5–7 days	IV

Supportive care:

- ☑ Monitoring of vital parameters and hemodynamic assessment
- ☑ Maintenance of the fluid-electrolyte balance
- ☑ Treatment of cardiovascular collapse
- ☑ Dialysis for renal failure

Prevention

- ☑ Rodent control
- ☑ Avoidance of contact with contaminated water and soil. Children should not wade through flood waters or play in puddles/stagnant water.
- ☑ Immunization of livestock (cattle, sheep, pigs, and horses) and family pets (cats and dogs).

Further Reading

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